

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT

Decode the Chart

Name: D. Hemasree

Roll No: 192524114

For each chart: What is it trying to tell you? Do you trust it? Why or why not?



Chart 1

Budget Allocation by Department (9 categories)



Chart 2

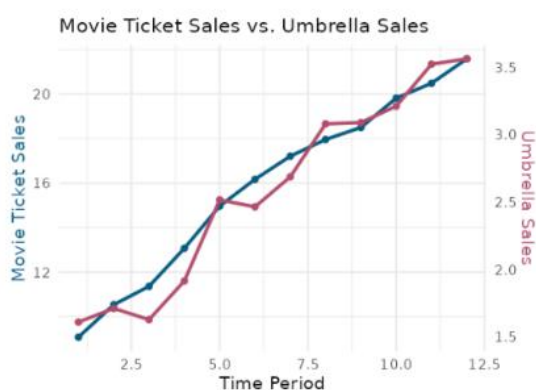


Chart 3

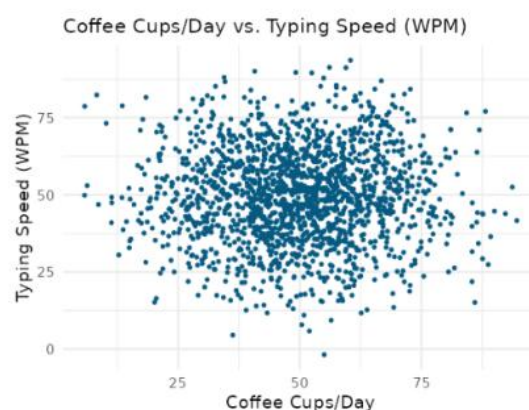


Chart 4

Your Notes:

Chart 1 — Regional Comparison (Bar Chart)

- **What it tells:** This chart compares attendance across four cities. It looks like Coimbatore is far ahead of the others and Madurai is far behind. At first glance, Coimbatore's bar looks about three times taller than Madurai's.

- **Do I trust it?** No, I do not trust it as it is. The chart gives a wrong first impression.
- **Why:** The y-axis starts at 620, not at 0. This is called a truncated y-axis. Because of this, the real values (about 651 for Coimbatore and 630 for Madurai) differ by only a few percent, but the bars make it look like a huge gap.
- **Fix:** A bar chart on a linear scale should always start at zero. This follows the Principle of Proportional Ink, where the bar size must match the real value. With a zero start, the bars would look almost equal.

Chart 2 — Budget Allocation by Department (Pie Chart)

- **What it tells:** This chart tries to show how the total budget is divided among 9 departments. Each slice is one department's share.
- **Do I trust it?** Not really. I cannot read it accurately, so I cannot use it to make a decision.
- **Why:** There are too many slices (9) and many colours look similar. Some slices are almost the same size, so nobody can say which one is the 3rd largest. Also, there are no labels on the slices, so we must keep checking a colour legend.
- **Fix:** Use a sorted bar chart, or write the labels directly on the chart. A pie chart works well only with about 6 categories or fewer.

Chart 3 — Movie Ticket Sales vs. Umbrella Sales (Dual Y-Axis)

- **What it tells:** This chart makes it look like movie ticket sales and umbrella sales go up together over time. The two lines follow each other very closely, so a viewer may think they are connected.
- **Do I trust it?** No. The link shown in this chart is false.
- **Why:** It uses two different y-axes, one on the left and one on the right, each with its own scale. The scales were chosen so the lines overlap. Movie tickets and umbrellas have no real connection, so this is a spurious (false) correlation. Correlation does not mean causation.
- **Fix:** Use one shared scale, or draw two separate charts side by side. We can also calculate a real correlation value instead of trusting the picture.

Chart 4 — Coffee Cups/Day vs. Typing Speed (Scatter Plot)

- **What it tells:** This chart tries to show whether drinking more coffee changes typing speed. Each dot is one person.
- **Do I trust it?** No. I cannot get a clear answer from this chart.
- **Why:** There are about 2,000 solid dots, and they overlap so much that they form one big blob. This is called overplotting. We cannot see where most of the data is concentrated, so we cannot tell if there is a real pattern or not.
- **Fix:** Make the dots partly transparent (alpha), or use a 2D histogram or hexbin chart. Contour lines also work. These show the dense areas clearly.

Your notes (Summary)

Charts can mislead even when the data is correct. Before trusting any chart, I should check four things: where the y-axis starts, how many

categories there are, whether two different axes are used, and whether the points overlap too much.